

AI-Powered Financial Document Question & Answer System

Using Retrieval-Augmented Generation (RAG)

NLP • Large Language Models • Prompt Engineering • Word Embeddings

Problem & Motivation

The Problem

- Financial documents (10-K, earnings reports, SEC filings) can exceed 300 pages.
- Analysts manually search dense documents for specific figures — slow and error-prone.
- Keyword search misses paraphrased content and synonyms.
- Raw LLMs without grounding hallucinate fabricated financial figures.

WHY IT MATTERS

300+

pages in a typical 10-K filing

\$5–15

cost per question if sent to GPT-4 directly

Banking & Finance

course-listed industry domain

Proposed Solution: Retrieval-Augmented Generation

Instead of asking the LLM everything, we let it look up answers in YOUR document.



Accurate

Answers come only from the actual document content.

No hallucinations.



Cited

Every answer includes the page number it came from.

Verifiable claims.



Efficient

Only relevant excerpts are sent to the LLM.

10× cheaper than full-doc.

Three AI Techniques (Course Requirement: Min 3)

1

Natural Language Processing

- Text preprocessing & cleaning
- Token-counted chunking (800 tok)
- Word2Vec (baseline)
- TF-IDF (baseline)
- Sentence-Transformers (proposed)

2

Large Language Models

- Groq Llama 3.3 70B (free) / GPT-4o-mini
- OpenAI-compatible Chat API
- Temperature = 0.1 (factual)
- Grounded answer generation
- Free local fallback mode

3

Prompt Engineering

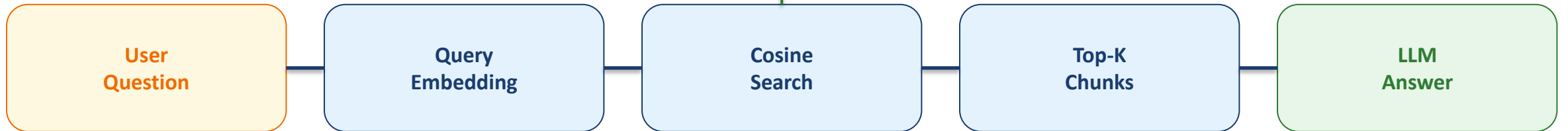
- Systematic 7-rule system prompt
- Chain-of-Thought (think step-by-step)
- Few-shot in-context learning
- Page citation enforcement
- Hallucination prevention

System Architecture: Two-Phase RAG Pipeline

PHASE 1 — INDEXING (runs once per document)



PHASE 2 — QUERYING (runs on every user question)



Result: when a user asks a question, the system retrieves the most relevant passages from the indexed document and generates a grounded answer with page citations — not from the LLM's training memory.

Implementation & Technology Stack

Layered Architecture

Web UI / API

Streamlit • FastAPI

Generation

Groq Llama 3.3 / GPT-4o-mini

Retrieval

ChromaDB • Cosine ANN

Embedding

MiniLM-L6-v2 (local, free)

NLP Preprocessing

pdfplumber • tiktoken

By the Numbers

28

Python source files

~2.4K

Lines of code

5

AI modules (ingest, retrieve, generate, eval, api)

4

REST API endpoints

3

Evaluation experiments + golden test cases

22

Pages: final LaTeX report

LIVE DEMO

Financial Document Q&A System

1. Upload a real 10-K filing (Apple FY 2023 from SEC EDGAR)
2. Watch the indexing pipeline: extract → chunk → embed → store
3. Ask: "What was the total net revenue in fiscal 2023?"
4. View the grounded answer with page citation and relevance score
5. Ask: "What are the main risk factors?" (multi-part answer)

Switch to browser: <http://localhost:8501>

Experiment 1: Embedding Baseline Comparison

50 real SEC 10-K Q&A pairs — does the method retrieve the exact source passage?

Method	Hit@1	MRR	Latency
TF-IDF (sparse, n-gram)	0.940	0.963	10.2 ms
Word2Vec (skip-gram, d=100)	0.640	0.717	4.3 ms
MiniLM-L6-v2 ← proposed	1.000	1.000	139 ms

KEY FINDING

1.000

Hit@1 = MRR = 1.000 (MiniLM-L6-v2)

*The exact source passage is
retrieved first for every query.*

*vs Hit@1 = 0.640 for Word2Vec
(TF-IDF a strong 0.940 baseline).*

Contextual sentence embeddings dramatically outperform classical NLP baselines for financial domain retrieval.

Experiment 2: RAG vs No-RAG — Why Retrieval Matters

50 financial-qa-10K questions, three answer strategies. Which approach stays grounded?

No-RAG

LLM alone, no document

Keyword Hit Rate	0.008
of gold keywords	<1%

Random Context

Irrelevant chunks fed in

Keyword Hit Rate	0.133
of gold keywords	13%

RAG (Proposed)

Correctly retrieved chunks

Keyword Hit Rate	0.840
of gold keywords	84%

RAG recovers 84% of gold-answer keywords versus under 1% without grounding — relevant retrieved context is what makes answers accurate.

Conclusions & Future Work

What We Achieved

- Built a production-ready RAG pipeline for financial documents
- Integrated 3 AI techniques: NLP, LLM, Prompt Engineering
- Evaluated on 50 real SEC 10-K Q&A pairs (financial-qa-10K)
- Perfect retrieval: Hit@1 = MRR = 1.000 for MiniLM
- RAG recovers 84% of gold keywords vs <1% without grounding
- Deployed live on Streamlit Cloud (free, Groq-powered)

Future Work

- Hybrid retrieval (dense semantic + BM25 sparse)
- FinBERT for domain-specific financial embeddings
- Cross-document queries (compare multiple companies)
- Cross-encoder re-ranking of top retrieved chunks
- Conversation memory for follow-up questions
- OCR support for scanned financial documents

The system satisfies all EC7203 requirements: 3 AI techniques, full evaluation with baselines, working demo, and reproducible code.

Thank You

Questions & Discussion

Deliverables Submitted:

Final Report (body under 20-page limit) • Working Web App • Source Code • This Video